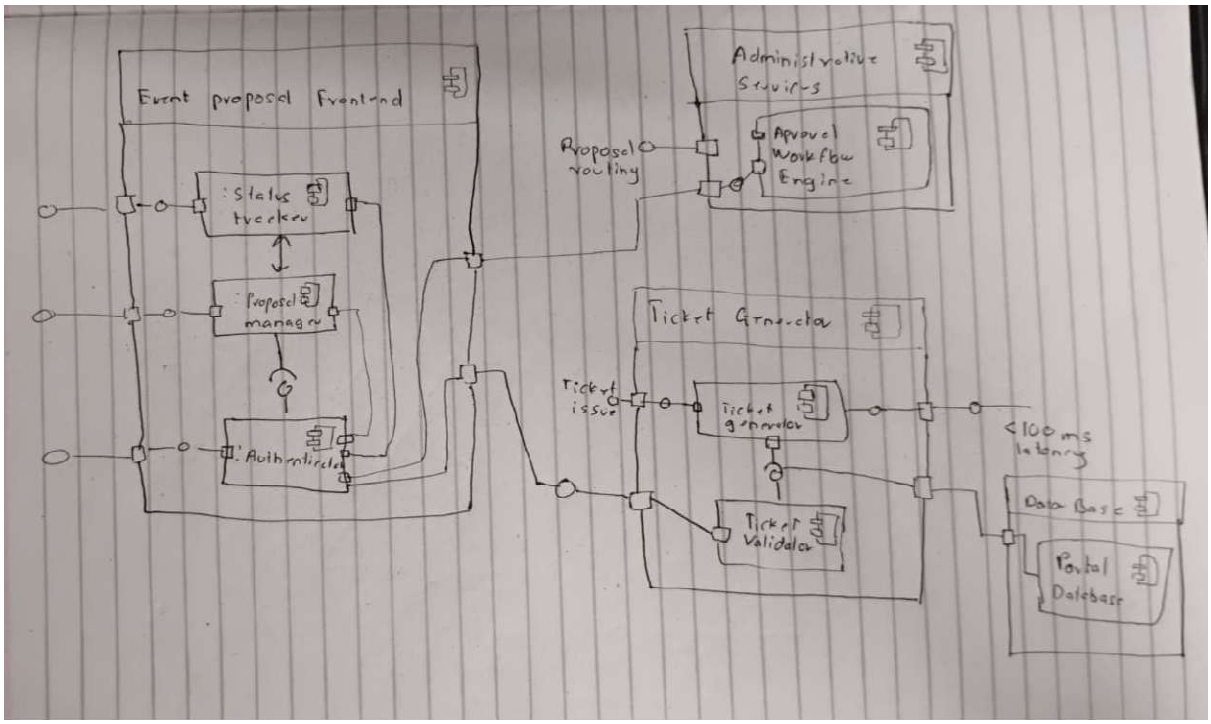
SRN :- PES2UG24CS446

Section :- H

Component Modelling




**Architecture Selection:** "We chose Microservices Architecture for the Student Club Event Ticketing & Budget Portal System."

**Architectural Choice** We selected the Microservices Architecture. This style divides the application into distinct, loosely coupled, and independently deployable subsystems (such as the Administrative Services for budgets and Ticketing & Access for event entry).

**Two Reasons**

1. **Separation of Distinct Workflows:** The system handles two completely different primary tasks: a multi-stage financial budget approval (Faculty to Finance to Dean) and real-time event ticketing. Using a microservices approach ensures that complex updates to the approval workflow engine do not interfere with the ticketing logic.
2. **Independent Scalability:** Event registration and ticket scanning often experience sudden traffic spikes right before or during an event. This architecture allows the TicketingServices component to scale independently to handle high user loads without needing to scale the entirely separate ApprovalWorkflowEngine.

**Security Advantage** This architecture naturally compartmentalizes sensitive operations, directly supporting the strict role-based access control requirement (NFR-002). By centralizing authentication into its own dedicated Authenticator component, the system ensures that an individual scanning tickets (Admin) cannot bypass internal network rules to access the multi-stage budget approval services (Dean/Finance), minimizing the attack surface.

**Performance Benefit** By decoupling the TicketingServices from heavy administrative database queries, the ticketing microservice can be highly optimized for speed. This isolation is critical for ensuring the system meets the high-priority performance requirement (NFR-001) of validating QR code scans in under 100 milliseconds during peak load, preventing long queues at the venue.